# Queue Management via Surveillance Camera for Patient Flow

by

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# Outline of Presentation


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# Background of the Study

## Introduction

In large healthcare facilities, such as hospitals, efficient management of patient flow is critical to improving service delivery and patient satisfaction. In most hospitals in Ghana, patient queues are managed using the First-Come-First-Served (FCFS) Taton et al. (2024) approach, which is largely manual and often prone to inefficiencies, errors, and unfair practices like queue jumping.

## Some effects of the manual method of managing queues

- Long waiting times
- Patient dissatisfaction
- Poor resource utilization
- Increased staff stress

# Background of the Study

This project proposes an automated queue monitoring and waiting time prediction system using computer vision. The system leverages existing surveillance cameras within hospital environments to capture video feeds, which are processed using computer vision models to detect, track, and analyse patient movement within queues.

## Automated Queue Management System

- Patient is assigned a unique identifier
- Continuous tracking of queue length
- Computing average waiting time
- Data collection for wait time prediction

The proposed system aims to reduce waiting times, improve fairness, and enhance the overall patient experience in healthcare facilities.

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# Problem Statement

Despite the high volume of patients in many Low- and Middle-Income Countries (LMICs), hospital queue management is largely handled manually using first-come, first-served (FCFS) method alongside other manual queue management methods. This manual approach to managing patient flow is prone to inefficiencies and errors. Because staff manage patients sequentially without real-time tracking tools, decisions about patient flow and resource allocation are often based on subjective judgment rather than objective data.

# Problem Statement

The absence of a centralized and automated queue monitoring system often results creates operational blindness which often result in;

- Difficulty in identifying peak periods
- Optimizing resource allocation
- Providing patients with accurate expectations regarding waiting times
- Depriving hospital management of historical data

Therefore, there is a critical need for an automated, intelligent queue monitoring system capable of providing real-time visibility, ensuring fairness, and supporting data-driven administration to improve patient flow and overall healthcare delivery.

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# Research Questions and Objectives

## Research questions

- How effectively can pre-trained object detection models be optimized for tracking patient movement in high-density hospital environments?
- To what extent can automated queue monitoring systems provide reliable real-time estimates of patient waiting times?
- How does the proposed system compare with manual queue management methods in terms of accuracy and fairness?
- How do environmental factors such as lighting conditions and camera placement affect the performance of computer vision-based queue monitoring systems?

# Research questions and objectives

## Objectives

- To utilize and adjust existing computer vision models (such as YOLO) for the real-time detection and anonymous tracking of patients.
- To integrate real-time tracking data with a mathematical model to estimate and predict patient waiting times.
- To conduct a comparative evaluation of the automated system against traditional manual methods, focusing on data integrity and fairness.
- To analyse and mitigate the impact of physical constraints, such as occlusion and variable lighting, on the system's counting accuracy

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# Methodology

- This study adopts a quantitative and experimental research design approach.
- The research leverages computer vision techniques integrated with mathematical modeling to transform passive surveillance infrastructure into an intelligent, real-time queue management tool.
- The quantitative framework focuses on measurable performance indicators: detection and tracking accuracy, waiting time prediction error (MAE/RMSE) and queue counting precision
- The research follows a two-stage systematic process:
  - System Design and Implementation
  - Testing/Evaluation

# System Design and Implementation

## 1. Data Acquisition Layer

- The system interfaces with existing hospital surveillance infrastructure via RTSP.
- OpenCV is then used to extract frames and continuously pass to the processing engine.

## 2. Computer Vision and Tracking Layer

- This layer serves as the “perceptual” core.
- It utilizes a YOLOv8 object detection model.
- Each detected person is assigned a unique temporal ID using ByteTrack tracking algorithm, allowing the system to monitor individual movements across frames

# System Design

## 3. Analytics and Prediction engine

- Use Point-in-Polygon test to detect queue membership in a defined ROI to tracked IDs.
- Calculate metrics from the queue: number of people in the queue, average waiting times
- The above metrics is then used to predict Expected Wait Time (EWT)

## 4. Presentation Layer

A Web-based Dashboard built with React that provides hospital administrators with live visualizations of queue density, average waiting times and EWT.

# Testing/Evaluation

- The evaluation will focus on three primary dimensions:
  - Detection and tracking accuracy
  - Queue metric accuracy
  - Waiting time prediction performance
- Each dimension will be measured against manually collected ground-truth data obtained by directly observing the same video footage.

## 1. Detection and Tracking Accuracy

This phase analyses how effective the YOLOV8 model identifies patient heads and how consistently the tracking algorithm maintains their unique IDs.

- **Metric:** Evaluation will include Precision, Recall, and F1-Score.
- **Validation:** Queue length will be manually counted from a 30 minute footage and compared to system logs.

# Testing/Evaluation

## 2. Queue Metric Accuracy

This dimension validates the spatial logic (the "Region of Interest"). The system must correctly distinguish between patients waiting in the queue and people simply walking past.

- **Metric:** Mean Absolute Percentage Error (MAPE) will be used to quantify the difference between the automated queue count and the actual count at 5-minute intervals.
- **Validation:** Using a "Point-in-Polygon" test, the system's ability to correctly classify individuals as "In-Queue" or "Out-of-Queue" will be verified against manual observations of the same video frames.

## 3. Waiting Time Prediction Performance

This is the core mathematical evaluation of the project. It measures the delta between the system's estimated wait time (EWT) and the actual time patients spent in the queue.

- **Metric:** Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n} |\text{Actual Wait}_i - \text{Predicted Wait}_i|$$

- **Validation:** The entry and exit timestamps for a sample size of [e.g., 50] patients will be manually recorded.The average of these manual durations (Ground Truth) will be compared against the system's predicted values to determine the model's reliability.

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# Justification

- Ghanaian hospitals experience severe overcrowding due to lack of automated queue systems.
- Manual monitoring is subjective and fails during peak demand periods.
- Traffic flow models (Fosu et al., 2020, 2021) offer a proven mathematical framework for queue dynamics.
- Existing surveillance infrastructure can be repurposed for non-intrusive, real-time monitoring.
- Optimisation methods Appati et al. (2015) enable efficient allocation of hospital resources.

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# Expected Outcome

## System Deliverables

- A real-time queue monitoring system with patient detection, counting, and waiting time prediction.
- Validated mathematical models for patient queue flow dynamics.
- Low prediction error (MAE/RMSE) and high detection accuracy.

## Broader Impact

- A scalable framework adaptable to other healthcare and public service settings.
- Data-driven insights to guide hospital staffing and queue administration.

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# Thank You